

Atharv Tiwari

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EDUCATION

GGSIU (Akhilesh Das Gupta Institute of Professional Studies)

Bachelor of Technology in Computer Science and Engineering

Aug 2024 – May 2028 (Expected)

New Delhi, India

PROJECTS

OKEANOS — Satellite Ocean Intelligence | *PyTorch, U-Net, FastAPI, AWS* (github.com/bugHunterYash/oceanos)

- Designed and trained a **U-Net that reconstructs ocean temperature at 15 depths** (surface to 1,000 m) from 7 daily satellite fields, with **29% lower thermocline error** than climatology on a held-out year.
- Engineered a data pipeline harmonising **5 satellite products and 13 years (4,748 days)** of reanalysis onto one grid, with leakage-free evaluation and validation against **ARGO float observations**.
- Benchmarked against climatology, ridge regression and gradient boosting on identical days over **three seeds**, with a full 15-level field inferred in **18 ms on a 4 GB laptop GPU**.

NetWM — Network Attack Forecasting | *PyTorch, Transformer, World Model (RSSM), FastAPI*

(github.com/ArunmehtaBuild/NetWM)

- Designed and trained a **Transformer + RSSM world model (~0.6M parameters)** that learns how a network's state evolves and forecasts compromise risk **five minutes ahead**, with the MITRE ATT&CK stage it is heading into.
- Reached **0.677 PR-AUC on a held-out day** against 0.139 for a logistic-regression baseline, working across **85M+ network flows** from CIC-IDS2017, CIC-IDS2018 and CTU-13.
- Generalised to an attack family never seen in training, with **0.96 ROC-AUC on brute-force attacks** against 0.45 for the baseline, and explained each forecast with attention and **Integrated Gradients**.

Onyx — AI-Powered DAST Platform | *Python, Security Tools, GitHub App API, Llama* (github.com/Atharv-tw/Onyx)

- Built a **high-concurrency vulnerability detection engine** integrating Subfinder, Nmap, and Nuclei with custom probes for **SQLi, SSRF, and IDOR** detection.
- Developed an **AI-driven triage system** using Llama (3.3 70b) models to analyze raw scanner outputs, achieving **90%+ accuracy** in deduplicating positives.
- Integrated automated remediation through the **GitHub App API**, generating verified security Pull Requests.

AWARDS & ACHIEVEMENTS

- eDC's Blueprint 6.0, IIT Delhi:** Competed till Delhi Regionals round. **Got invited to Emergence, IIT Delhi's** Flagship incubation program as well as got **BeCon'26** special invitations.
- IEEE T-Hacks 8.0: Winner** among **800+ participants** in a 24-hour hackathon.
- Shloka Decode 2.0, NSUT:** Emerged **Winner** among 500+ participants in an 8-hour hackathon.

EXPERIENCE

Nexera (Nexeraofficial.in)

Nov 2025 – Present

Founding Chief Technical Officer

Remote

- Led the technical development of a **production platform serving 3K+ daily users**, owning architecture, infrastructure, and key product decisions.
- Engineered **secure premium-content delivery** using Redis-based concurrent-stream protection and access-control workflows; built a centralized admin panel with **role-based access control** for premium content management.

Adxplorers

Jun 2026 – Jul 2026

Full-Stack AI Intern

Remote

- Designed and implemented **evaluation datasets and test cases for multiple LLMs**, assessing model responses across defined quality and performance criteria.
- Analyzed evaluation results to improve the **reliability and quality of model outputs**.

AAIRO (AI & Robotics Society of ADGIPS)

Sep 2026 – Present

Computational Intelligence Lead

ADGIPS

- Lead the society's **AI, ML and data science domains**.
- Building the **intelligence pipeline for a damage-assessment robot**.

TECHNICAL SKILLS

Languages: Python, SQL, C, JavaScript

Frameworks: Next.js, Node.js, FastAPI

AI/ML: PyTorch, scikit-learn, XGBoost, NumPy, pandas, Transformers, U-Net, RAG, AI Agents, Vector Databases, Llama, Claude

Systems / Infrastructure: Redis, Docker, Concurrent Systems, Job Queues, API Architecture, Secure Content Delivery.